



STTHK 3133

SENSOR-BASED SYSTEMS

GROUP A

MIDTERM

SEMESTER A242

TITLE:

**TEMPERATURE AND HUMIDITY MONITORING WITH RELAY TRIGGER
AND NEAR REAL-TIME GRAPH**

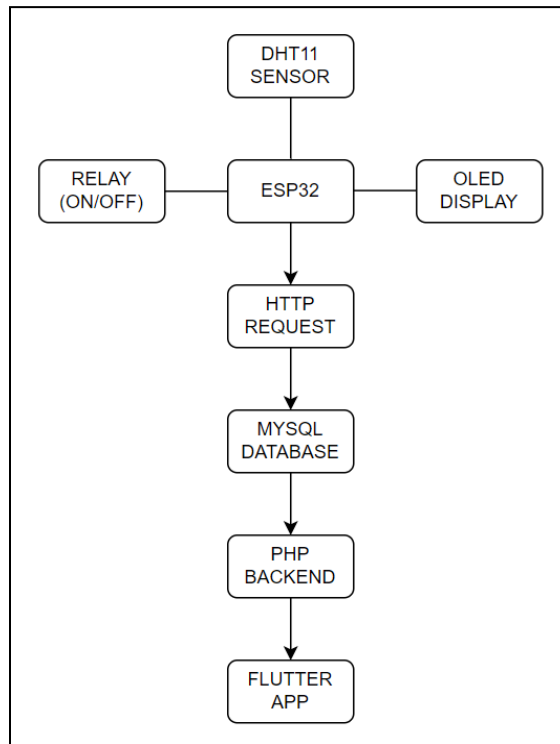
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1.0 System architecture diagram



**Figure 1: Temperature and humidity monitoring and control system
architecture diagram**

- Temperature and humidity are measured by the DHT11 sensor.
- Information is displayed on an OLED screen by the ESP32 microcontroller, which also gathers data from the sensor and manages a relay (for example, to turn a device on or off).
- The ESP32 uses an HTTP request to transfer sensor data to a distant server.
- A MySQL database holds the data.
- Data processing and client data delivery are handled via a PHP backend.
- Viewing sensor data and remotely controlling the system are done through a Flutter app.

2.0 Setup steps

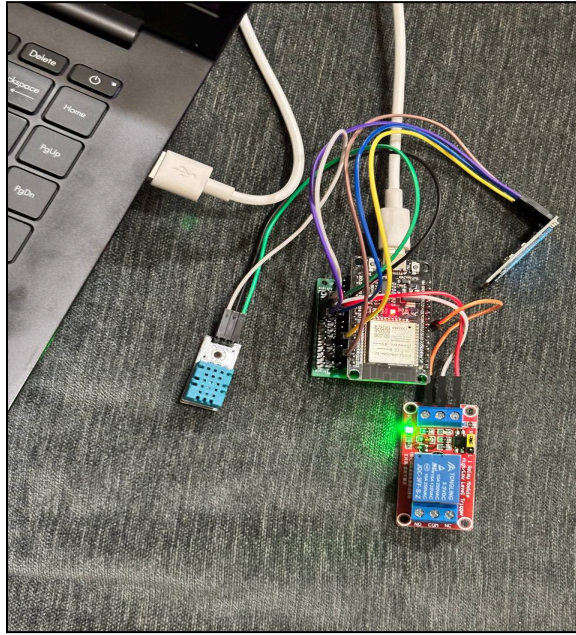


Figure 2: Hardware setups

1. Hardware Setup

- Connect the ESP32 as the main controller.
- Attach the DHT11 sensor to the ESP32 to measure temperature and humidity.
- Connect the Relay module to the ESP32 to trigger when sensor readings exceed limits.
- Connect the OLED display to show live temperature and humidity readings.

2. Backend Setup

- Set up the backend framework (PHP) and database (MySQL).
- Create an API endpoint to receive sensor data from ESP32 every 10 seconds.
- Implement logic to save sensor readings with timestamps into the database.
- Create an API endpoint that allows the mobile/web app to fetch stored data for graphing.

3. Mobile App/Web Setup

- Develop the app using Flutter.
- Connect the app to the backend API to fetch temperature and humidity data.
- Implement a graph/chart to visually display the fetched data for monitoring.

3.0 Screenshots (backend, relay, app, OLED)

				id	device_id	temperature	humidity	relay_state	timestamp	create_at
☐	Edit	Copy	Delete	1	1	31	67	1	1110025	2025-05-30 05:16:46
☐	Edit	Copy	Delete	2	1	31	67	1	1120025	2025-05-30 05:16:56
☐	Edit	Copy	Delete	3	1	30.9	67	1	1130026	2025-05-30 05:17:06
☐	Edit	Copy	Delete	4	1	31	67	1	1140025	2025-05-30 05:17:16
☐	Edit	Copy	Delete	5	1	31	67	1	1150025	2025-05-30 05:17:26
☐	Edit	Copy	Delete	6	1	31	67	1	1160025	2025-05-30 05:17:36
☐	Edit	Copy	Delete	7	1	31	67	1	1170025	2025-05-30 05:17:46
☐	Edit	Copy	Delete	8	1	31	67	1	1180025	2025-05-30 05:17:56
☐	Edit	Copy	Delete	9	1	31	67	1	1190025	2025-05-30 05:18:06
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☐	Edit	Copy	Delete	17	1	31	67	1	1270025	2025-05-30 05:19:26

Figure 3: Backend - tbl_sensordata in phpMyAdmin

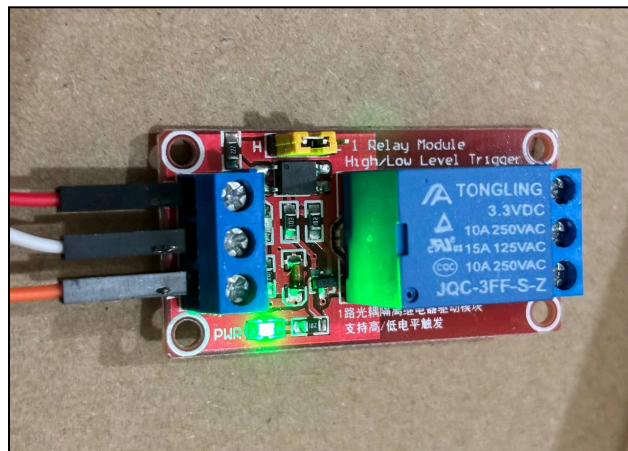


Figure 4: Relay LED (ON/OFF)

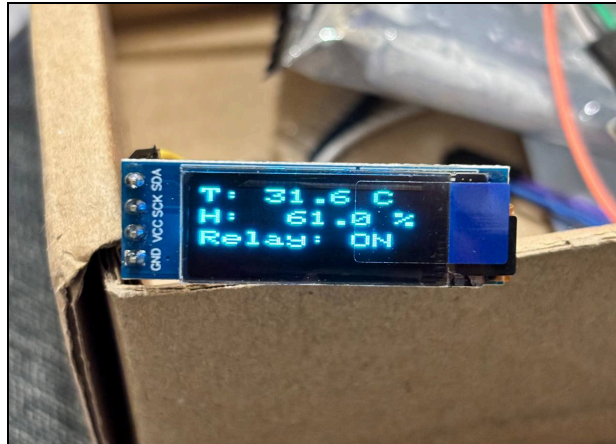


Figure 5: Real-time OLED display

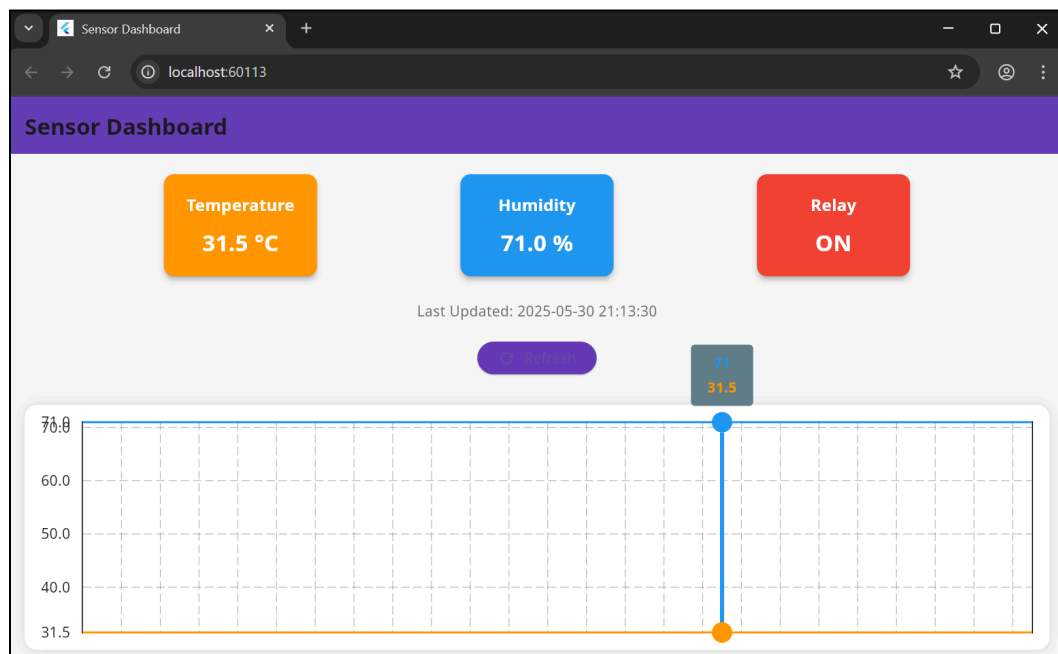


Figure 6: Flutter web-based application

- showing real-time Temperature, Humidity, Relay Status

4.0 Challenges and improvements

The biggest issue I ran into with this project was time management since ultimately it took me far more than 48 hours of actual development to complete. Another factor that greatly complicated this process, was that I am a beginner with Flutter, and this was my very first experience using this framework, especially since I had not studied it in the semester before and was not at all familiar with the software. I also had several technical issues with Flutter installation which involved uninstalling and reinstalling the software more than 5 times due to problems with missing files, and the installations being incomplete. This whole process was very time consuming and contributed to my delays, which ultimately resulted in me being unable to finish the project in the time originally allocated.

Despite my complications with time management and installation, I was able to finish the hardware configuration strand and backend development within the time allocated. There was also a considerable problem with connection between the Arduino code and the database, in that both POST and GET requests had to connect with the correct insert and retrieval functions, so I had to ensure the data flow did not break down. While I was unable to accomplish most of the requirements on the original schedule, I tried my best to complete the project, and submit it as outlined.

One important improvement plan would be better time management and planning. Assigning realistic time allocated to tasks in each part of the project would be ready for unpredictable delays, which are bound to happen while working with unfamiliar technologies. When doing the planning, some internal time can be included for troubleshooting and testing to address problems, which I think can greatly improve overall effectiveness while ensuring deadlines are met.

Another improvement plan would be having peer or mentor support. Working with people who have used Flutter or have built IoT projects can add valuable insight and allow for troubleshooting issues quicker and improving the overall timeliness and quality of the work and product. I did ask some classmates even though there were some delays in answering my doubtness but they still answered it, but I truly understood we were working on the same tasks.

Link(s):

[Youtube Demonstration](#)